An outbreak of neurological autoimmunity with polyradiculoneuropathy in workers exposed to aerosolised porcine neural tissue: a descriptive study.

BACKGROUND: Between November, 2006, and May, 2008, a subacute neurological syndrome affected workers from two swine abattoirs in Minnesota and Indiana who had occupational exposure to aerosolised porcine brain. We aimed to describe the pathogenic and immunological characteristics of this illness. METHODS: All patients from two abattoirs who presented or were referred to the Mayo Clinic (Rochester, MN, USA) with neurological symptoms were included. We recorded details of exposure to aerosolised brain tissue and did comprehensive neurological, laboratory, neuroimaging, electrophysiological, pathological, and autoimmune serological assessments. Healthy controls were recruited from the community and from workers at the plant in Minnesota. FINDINGS: 24 patients were identified (21 from Minnesota, three from Indiana). The shortest duration from first exposure to symptom onset was 4 weeks. No infectious agent that could trigger disease was identified. All patients developed polyradiculoneuropathy, which was usually sensory predominant and painful. Two patients had initial CNS manifestations: transverse myelitis and meningoencephalitis. Nerve conduction studies localised abnormalities to the most proximal and distal nerve segments. Quantitative sensory and autonomic testing revealed involvement of large and small sensory fibres and sweat fibres. MRI showed prominent abnormalities of roots and ganglia. Nerve biopsies identified mild demyelination, axonal degeneration, and perivascular inflammation. Protein concentrations were high in the CSF of 18 (86%) of 21 patients. Sera from all patients and 29 (34%) of 85 unaffected workplace controls (but none of 178 community controls) had a distinctive neural-reactive IgG; 75% of patients' sera contained an IgG specific to myelin basic protein. Seropositivity correlated directly with exposure risk in patients and controls. 17 patients required immunomodulatory therapies, six improved spontaneously, and one was lost to follow-up after exposure stopped. INTERPRETATION: The neurological disorder described is autoimmune in origin and is related to occupational exposure to multiple aerosolised porcine brain tissue antigens. The pattern of nerve involvement suggests vulnerability of nerve roots and terminals where the blood-nerve barrier is most permeable. FUNDING: Mayo Clinic Foundation; Minnesota Department of Health; Centers for Disease Control and Prevention. Copyright 2010 Elsevier Ltd. All rights reserved.